REPUBLIC OF YEMEN: EXTERNAL AND PUBLIC DEBT SUSTAINABILITY ANALYSIS (DSA)

1. This note examines the sensitivity of Yemen's external and public debt dynamics under the staff's baseline scenario to a number of standardized shocks in the debt sustainability framework for low-income countries. The baseline scenario assumes that the government implements reform measures such as the elimination of the fuel subsidy, the introduction of GST, or undertakes structural measures to improve the business environment and enhance non-oil sector growth the medium and long term.

A. The Current Situation

2. Yemen's total external public debt was $5.1 billion at end-2004 (39 percent of GDP), of which $1.3 billion is owed to the Russian Federation. Other external debts (46 percent of total external debt) are mostly from multilateral creditors, primarily the World Bank, at highly concessional terms. Domestic public debt, all denominated in local currency, accounted for about 5.6 percent of GDP at end-2004.

3. The baseline scenario underlying the DSA is based on the medium-term framework under the adjustment scenario in the staff report. It assumes that future borrowings to meet the external financing requirements are made at a highly concessional rate given the impending vulnerabilities of the economy.

B. Public Debt Sustainability

4. Under the baseline scenario, although the total public sector debt starts at acceptable levels in 2004 (45 percent of GDP, see Table 1), it increases over the medium term to reach 69.9 percent in 2009. This is mainly due (as discussed before) to the tapering of revenues and grants with more than 10 percent of GDP from 2004 to 2009. Public sector debt stocks in NPV terms start at 27.4 percent in 2004, but rise rapidly to 51.1 percent of GDP in 2009. The NPV of the debt-to-revenue ratio is projected to rise from 78.2 percent in 2004 to 232 percent in 2009 and to 804 percent in 2023.

5. Stress tests (Table 2 and Figure 1) show that Yemen's fiscal sustainability will depend on the implementation of actions to sustain and improve fiscal adjustment and growth, particularly in the non-oil economy. The main results are the following:

6. Given the fact that Yemen was able to substantially improve its fiscal position during the second half of the 1990s,[23] it is possible under scenario (1), that assumes real GDP growth and the primary balance at historical (low) averages, to improve the NPV of debt-to-

[23] Improvement as measured in the change in the primary balance from a deficit of 11.4 percent of GDP in 1994 to a primary surplus of 4.4 percent in 2001.

GDP to 36 percent in 2009,[24] compared to 51 percent in the baseline scenario. However, in 2009 the debt service-to-revenue ratio would be almost 10 percent lower than in the baseline scenario. Under scenario (2), where the primary balance is kept unchanged from 2003 onwards to "historic averages" scenario mentioned above, the NPV of debt-to-GDP would increase to 49 percent and the debt-to-revenue ratio to 222 percent in 2009.

7. Scenarios (1) and (2) discussed above are however not feasible given: (a) the dependency of public finances on oil revenues; and (b) the expected depletion of oil reserves by 2016.

8. A more realistic stress would be to impose a shock to real growth (lowering it permanently by reducing it to the historical average minus one standard deviation). Under this scenario a significant deterioration in all debt ratios would take place. In 2009, the NPV of debt-to-GDP would reach 55 percent; the NPV of debt-to-revenue would increase to 251 percent; and debt service would absorb 25 percent of revenue.

C. External Debt Sustainability

9. Under the baseline scenario, although the external debt stock in NPV terms starts at a modest level (22 percent of GDP in 2004) and remains there over the medium term (35 percent in 2013), debt indicators measuring Yemen's ability to service external debt deteriorate significantly over the long term as Yemen runs out of oil. The NPV of the debt-to-exports would increase to about 340 percent, and the debt service-to-exports ratio to 20 percent by 2023. The risks to the debt dynamics primarily stem from the fact that oil, the largest source of export earnings for Yemen, is expected to virtually run out by 2016, generating a spike in sustainability indicators. Subsequently, debt indicators improve with the growth in non-oil exports.

10. External debt dynamics are most vulnerable to the decline in exports. A shock to export growth equal to the size of one standard deviation under the stress tests (B2) raises the NPV of debt-to-exports ratio to about 560, and debt service would absorb 36 percent of exports in 2013. This scenario clearly demonstrates that, without a strong commitment to export diversification, Yemen's external debt would be unsustainable.

11. A lack of concessional new borrowing will also aggravate the external sustainability by pushing up the NPV-to-export ratio to over 270 by 2013.[25]

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[24] The public debt-to-GDP ratio in 2009 would be at 69.9 percent of GDP under the baseline scenario (see Table 1).

[25] Given that growth and capital account surplus were high in the past, primarily reflecting the performance of the oil-economy, the scenario with historical averages delivers implausible debt ratios.

Table 1. Republic of Yemen: Public Sector Debt Sustainability Framework, Baseline Scenario, 2000–23

(In percent of GDP, unless otherwise indicated)

	Actual				Historical	Standard	Projections						2004–09			2009–23
	2000	2001	2002	2003	Average 5/	Deviation 5/	2004	2005	2006	2007	2008	2009	Average	2013	2023	Average
Public sector debt 1/	**58.8**	**57.8**	**52.8**	**53.1**	**103.0**	**50.5**	**45.1**	**45.8**	**50.4**	**56.0**	**62.5**	**69.9**	**54.9**	**120.2**	**258.4**	**161.8**
Of which: Foreign-currency denominated	51.5	52.5	49.5	45.1			39.5	37.4	38.0	38.4	38.6	39.2		75.6	187.8	
Change in public sector debt	-37.9	-1.0	-5.0	0.3			-8.0	0.7	4.6	5.6	6.4	7.4		10.6	16.1	
Identified debt-creating flows	-34.0	-2.9	-2.0	-0.8			-4.8	-0.4	3.3	5.1	5.8	7.3		11.4	16.1	
Primary deficit	**-10.3**	**-4.7**	**-0.8**	**3.3**	**1.4**	**6.7**	**1.3**	**1.4**	**3.6**	**4.9**	**6.4**	**6.9**	**4.1**	**8.1**	**9.2**	**8.4**
Revenue and grants	39.2	35.3	33.6	33.1			35.0	30.6	27.3	24.7	22.8	21.9		19.9	19.0	
Of which : Grants	1.3	0.3	1.6	0.4			0.2	0.8	0.8	0.8	0.8	0.8		1.0	2.2	
Primary (noninterest) expenditure	28.9	30.6	32.8	36.4			36.3	32.1	30.9	29.5	29.2	28.9		28.0	28.3	
Automatic debt dynamics	-19.2	2.4	-1.2	-4.1			-5.5	-1.8	-0.4	0.2	-0.6	0.3		3.3	6.9	
Contribution from interest rate/growth differential	-18.8	2.0	-0.8	-4.0			-5.0	-2.7	0.0	0.1	-0.6	-0.2		3.3	6.9	
Of which : Contribution from average real interest rate	-14.7	4.6	1.4	-2.4			-3.6	-1.5	1.1	1.4	1.1	1.6		6.7	16.1	
Of which : Contribution from real GDP growth	-4.1	-2.6	-2.2	-1.6			-1.4	-1.3	-1.2	-1.3	-1.7	-1.8		-3.4	-9.2	
Contribution from real exchange rate depreciation	-0.4	0.4	-0.4	-0.1			-0.5	0.9	-0.3	0.2	0.0	0.5		...	...	
Denominator = 1+g	1.0	1.0	1.0	1.0			1.0	1.0	1.0	1.0	1.0	1.0		1.0	1.0	
Other identified debt-creating flows	-4.5	-0.6	0.0	0.0			-0.6	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-4.5	-0.6	0.0	0.0			-0.6	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-3.9	1.8	-3.0	1.1			-3.2	1.1	1.4	0.5	0.6	0.2		-0.9	0.0	
NPV of public sector debt	**32.2**	**30.8**	**29.0**	**31.7**			**27.4**	**28.4**	**32.8**	**38.1**	**44.4**	**51.1**		**77.7**	**153.0**	
Of which: Foreign-currency denominated	24.9	25.5	25.7	23.7			21.8	20.1	20.4	20.5	20.5	20.4		33.1	82.3	
Of which: External	24.9	25.5	25.7	23.7			21.8	20.1	20.4	20.5	20.5	20.4		33.1	82.3	
NPV of contingent liabilities (not included in public sector debt)	...	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	73.2	46.8	51.8	48.7			51.7	42.2	48.2	54.2	60.5	67.2		113.8	237.2	
NPV of public sector debt-to-revenue ratio (in percent) 3/	82.2	87.1	86.4	95.8			78.2	92.8	120.1	154.5	194.3	232.7		390.4	804.1	
Of which: External	63.5	72.1	76.6	71.8			62.2	65.6	74.7	83.3	89.9	93.0		166.3	432.8	
Debt service-to-revenue ratio (in percent) 3/ 4/	35.0	15.9	13.5	8.0			11.7	10.2	13.2	16.4	20.4	23.7		35.5	67.7	
Primary deficit that stabilizes the debt-to-GDP ratio	27.6	-3.7	4.2	3.0			9.3	0.7	-1.0	-0.8	-0.1	-0.5		-2.5	-6.8	
Key macroeconomic and fiscal assumptions																
Real GDP growth (in percent)	4.4	4.6	3.9	3.1	5.6	2.7	2.7	2.9	2.7	2.7	3.0	3.0	2.8	3.2	3.9	3.4
Average nominal interest rate on forex debt (in percent)	0.9	1.1	1.0	1.2	1.6	1.1	1.2	1.2	1.1	1.1	1.2	1.1	1.1	0.5	0.1	0.5
Average real interest rate on domestic currency debt (in percent)	-12.0	22.8	24.7	31.7	-1.5	22.1	9.1	28.1	22.9	16.9	11.8	9.1	16.3	7.3	7.4	7.4
Real exchange rate depreciation (in percent, + indicates depreciation)	-0.7	0.8	-0.8	-0.2	26.9	70.2	...	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	25.7	-0.1	4.9	12.5	18.4	16.6	14.4	13.4	8.3	7.6	8.9	9.0	10.3	6.6	7.0	7.0
Growth of real primary spending (deflated by GDP deflator, in percent)	19.1	10.8	11.2	14.4	10.3	20.0	2.4	-9.0	-0.9	-2.0	2.0	1.7	-1.0	2.7	7.4	3.2
Grant element of new external borrowing (in percent)	0.6	0.6	0.6	0.6	0.6	0.0	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6	...

Sources: Country authorities; and Fund staff estimates and projections.

1/ Central government debt (net).

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2. Republic of Yemen: Sensitivity Analyses for Key Indicators of Public Sector Debt , 2003–2023

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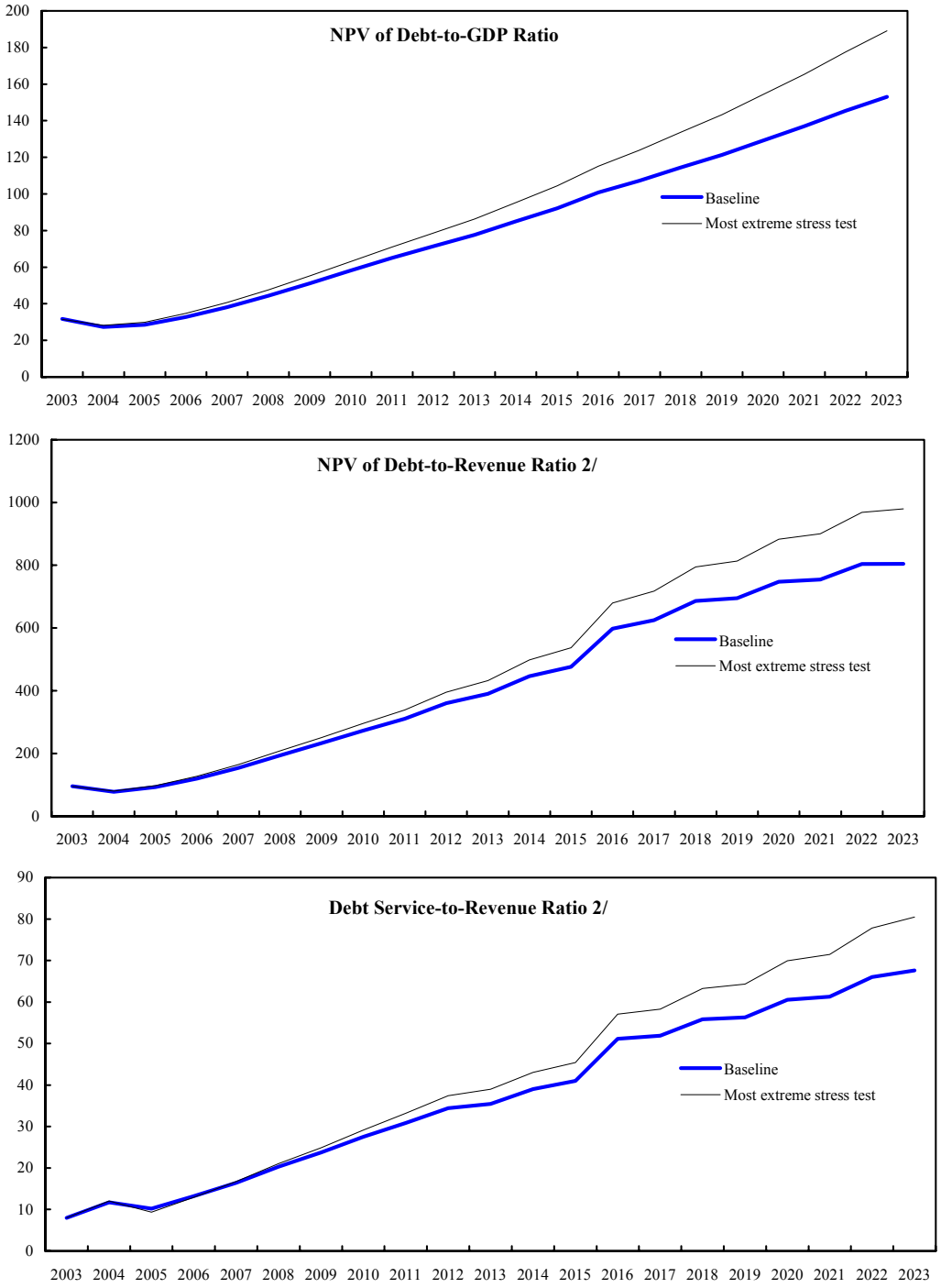
	Actual	Projections										
	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2023
NPV of Debt-to-GDP Ratio												
Baseline	**32**	**27**	**28**	**33**	**38**	**44**	**51**	**58**	**65**	**71**	**78**	**153**
A. Alternative scenarios												
A1. Real GDP growth and primary balance are at historical averages	31	27	27	29	31	34	36	37	38	38	38	31
A2. Primary balance is unchanged from 2003	31	29	31	35	40	44	49	53	57	60	63	84
A3. Permanently lower GDP growth 1/	31	28	30	35	41	48	55	63	71	79	86	189
B. Bound tests												
B1. Real GDP growth is at historical average minus one standard deviations in 2004–05	31	28	29	33	38	44	51	57	64	70	76	146
B2. Primary balance is at historical average minus one standard deviations in 2004–05	31	32	37	41	46	52	58	65	72	79	86	164
B3. Combination of B1-B2 using one half standard deviation shocks	31	30	32	36	41	47	54	61	67	74	80	157
B4. One time 30 percent real depreciation in 2004	31	38	37	41	45	51	57	63	67	71	77	161
B5. 10 percent of GDP increase in other debt-creating flows in 2004	31	34	34	39	44	50	56	63	70	77	83	160
NPV of Debt-to-Revenue Ratio 2/												
Baseline	**96**	**78**	**93**	**120**	**155**	**194**	**233**	**273**	**311**	**360**	**390**	**804**
A. Alternative scenarios												
A1. Real GDP growth and primary balance are at historical averages	95	77	90	107	127	147	163	175	184	195	195	169
A2. Primary balance is unchanged from 2003	95	83	102	130	161	193	222	249	273	304	317	440
A3. Permanently lower GDP growth 1/	95	80	97	127	165	208	251	296	339	396	432	980
B. Bound tests												
B1. Real GDP growth is at historical average minus one standard deviations in 2004–05	95	79	94	121	155	194	231	269	305	352	380	771
B2. Primary balance is at historical average minus one standard deviations in 2004–05	95	92	120	148	185	226	265	307	346	399	430	862
B3. Combination of B1-B2 using one half standard deviation shocks	95	84	104	132	167	207	245	285	322	373	403	827
B4. One time 30 percent real depreciation in 2004	95	107	122	150	184	223	259	294	323	361	389	848
B5. 10 percent of GDP increase in other debt-creating flows in 2004	95	97	112	141	177	218	257	298	336	388	418	841
Debt Service-to-Revenue Ratio 2/												
Baseline	**8**	**12**	**10**	**13**	**16**	**20**	**24**	**28**	**31**	**34**	**35**	**68**
A. Alternative scenarios												
A1. Real GDP growth and primary balance are at historical averages	8	12	9	11	12	14	15	17	18	19	19	21
A2. Primary balance is unchanged from 2003	8	12	10	14	16	19	20	23	25	28	28	44
A3. Permanently lower GDP growth 1/	8	12	9	13	17	21	25	29	33	37	39	80
B. Bound tests												
B1. Real GDP growth is at historical average minus one standard deviations in 2004–05	8	12	9	12	16	20	23	27	30	34	35	65
B2. Primary balance is at historical average minus one standard deviations in 2004–05	8	12	13	18	18	21	24	28	32	35	36	72
B3. Combination of B1-B2 using one half standard deviation shocks	8	12	11	15	16	20	23	27	30	34	35	69
B4. One time 30 percent real depreciation in 2004	8	12	10	13	17	21	24	28	32	35	36	68
B5. 10 percent of GDP increase in other debt-creating flows in 2004	8	12	15	14	17	21	24	28	31	35	36	70

Sources: Country authorities; and Fund staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).

2/ Revenues are defined inclusive of grants.

Figure 1. Repblic of Yemen: Indicators of Public Debt Under Alternative Scenarios, 2003–2023 1/
(In percent)



Source: Staff projections and simulations.

1/ Most extreme stress test is test that yields highest ratio in 2013.
2/ Revenue including grants.

Table 3. Republic of Yemen: External Debt Sustainability Framework, Baseline Scenario, 2000–2023 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical	Standard	Est.	Projections									
													2003–09			2010–23
	2000	2001	2002	Average 6/	Deviation 6/	2003	2004	2005	2006	2007	2008	2009	Average	2013	2023	Average
External debt (nominal) 1/	**50.5**	**51.1**	**48.6**			**44.4**	**39.3**	**36.4**	**36.3**	**36.4**	**36.3**	**36.5**	**37.9**	**64.9**	**168.1**	**101.0**
Of which: Public and publicly guaranteed (PPG)	50.5	51.1	48.6			44.4	39.3	36.4	36.3	36.4	36.3	36.5	37.9	64.9	168.1	101.0
Change in external debt	-22.4	0.5	-2.5			-4.2	-5.1	-3.0	-0.1	0.1	-0.1	0.3	-1.7	8.6	11.4	9.4
Identified net debt-creating flows	-30.3	-7.8	-10.7			-4.1	-5.8	-0.7	2.3	4.9	6.2	7.5	1.5	5.7	0.3	4.3
Non-interest current account deficit	**-13.8**	**-5.9**	**-5.8**	**-5.0**	**4.3**	**-1.6**	**-1.4**	**-0.1**	**3.7**	**6.8**	**8.8**	**10.1**	**3.7**	**10.5**	**12.1**	**11.7**
Deficit in balance of goods and services	-6.4	1.1	0.7			1.5	0.3	2.4	7.9	12.2	14.9	16.6	8.0	19.9	31.1	24.3
Exports	41.6	36.0	38.4			37.3	37.3	35.7	29.9	25.2	22.0	20.7	29.7	17.6	26.2	18.9
Imports	35.2	37.2	39.1			38.8	37.6	38.1	37.8	37.4	37.0	37.3	37.7	37.5	57.3	43.3
Net current transfers (negative = inflow)	-14.6	-13.4	-13.9	-17.9	4.0	-10.7	-9.9	-8.5	-8.2	-7.8	-7.4	-7.2	-8.5	-9.0	-17.9	-12.0
Other current account flows (negative = net inflow)	7.2	6.3	7.3			7.6	8.2	5.9	4.0	2.4	1.3	0.6	4.3	-0.4	-1.0	-0.6
Net FDI (negative = inflow)	**-1.5**	**-2.7**	**-3.0**	**-0.6**	**2.5**	**-1.6**	**-3.8**	**0.0**	**-0.9**	**-1.4**	**-2.1**	**-2.1**	**-1.7**	**-3.8**	**-7.2**	**-5.3**
Endogenous debt dynamics 2/	**-15.0**	**0.7**	**-1.8**			**-0.9**	**-0.6**	**-0.6**	**-0.5**	**-0.4**	**-0.5**	**-0.5**	**-0.5**	**-1.0**	**-4.6**	**-2.2**
Contribution from nominal interest rate	0.5	0.6	0.5			0.5	0.4	0.4	0.5	0.5	0.6	0.6	0.5	0.9	1.7	1.2
Contribution from real GDP growth	-2.5	-2.3	-1.9			-1.4	-1.0	-1.0	-0.9	-0.9	-1.0	-1.0	-1.0	-1.8	-6.3	-3.4
Contribution from price and exchange rate changes	-13.0	2.5	-0.4			…	…	…	…	…	…	…	…	…	…	…
Residual (3-4) 3/	**7.9**	**8.4**	**8.2**			**-0.1**	**0.7**	**-2.2**	**-2.4**	**-4.8**	**-6.3**	**-7.2**	**-3.2**	**2.9**	**11.0**	5.1
Of which: Exceptional financing	-0.6	0.0	0.0			0.0	-0.6	0.0	0.0	0.0	0.0	0.0	-0.1	0.0	0.0	0.0
NPV of external debt 4/	...	...	26.1			24.1	21.9	20.7	20.8	21.0	21.0	21.2	21.5	35.1	88.1	53.6
In percent of exports	...	...	68.0			64.5	58.6	57.9	69.6	83.4	95.4	102.6	76.0	199.4	335.8	282.7
NPV of PPG external debt	**...**	**...**	**26.1**			**24.1**	**21.9**	**20.7**	**20.8**	**21.0**	**21.0**	**21.2**	**21.5**	**35.1**	**88.1**	**53.6**
In percent of exports	...	...	68.0			64.5	58.6	57.9	69.6	83.4	95.4	102.6	76.0	199.4	335.8	282.7
Debt service-to-exports ratio (in percent)	6.1	6.4	3.8			3.4	3.4	2.8	4.0	5.0	6.3	6.6	4.5	11.6	19.7	16.2
PPG debt service-to-exports ratio (in percent)	6.1	6.4	3.8			3.4	3.4	2.8	4.0	5.0	6.3	6.6	4.5	11.6	19.7	16.2
Total gross financing need (billions of U.S. dollars)	-1.2	-0.6	-0.7			-0.2	-0.5	0.1	0.6	1.1	1.4	1.7	0.6	1.5	1.3	1.5
Non-interest current account deficit that stabilizes debt ratio	8.6	-6.4	-3.3			2.5	3.7	2.9	3.8	6.7	8.9	9.8	5.5	1.9	0.8	2.3
Key macroeconomic assumptions																
Real GDP growth (in percent)	4.4	4.6	3.9	5.1	1.4	3.1	2.7	2.9	2.7	2.7	3.0	3.0	2.8	3.2	3.9	3.4
GDP deflator in US dollar terms (change in percent)	21.6	-4.6	0.8	8.1	16.9	8.9	12.1	9.7	3.2	3.0	3.7	2.4	6.8	-4.8	-6.1	-5.1
Effective interest rate (percent) 5/	0.9	1.1	1.0	1.1	0.2	1.2	1.2	1.2	1.3	1.4	1.7	1.6	1.3	1.5	1.1	1.3
Growth of exports of G&S (U.S. dollar terms, in percent)	53.4	-13.7	11.5	13.1	32.0	9.2	15.1	8.1	-11.3	-10.9	-6.6	-1.1	0.6	-3.7	6.8	-0.2
Growth of imports of G&S (U.S. dollar terms, in percent)	9.6	5.2	10.1	6.7	6.6	11.4	11.5	14.6	5.0	4.5	5.7	6.5	8.8	-1.6	3.2	1.0
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	36.4	31.6	35.5	42.6	42.1	43.4	42.6	38.6	51.8	51.5	50.5
Memorandym item:																
Nominal GDP (billions of U.S. dollars)	9.6	9.5	10.0			11.2	12.9	14.6	15.4	16.3	17.4	18.4		17.2	13.1	

Source: Staff simulations.

1/ Includes both public and private sector external debt.

2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that NPV of private sector debt is equivalent to its face value.

5/ Current-year interest payments devided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Republic of Yemen: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2003-23

(In percent)

	Est.	Projections						
	2003	2004	2005	2006	2007	2008	2013	2023
NPV of debt-to-GDP ratio								
Baseline	24	22	21	21	21	21	35	88
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2004-23 1/	24	22	18	12	6	0	0	0
A2. New public sector loans on less favorable terms in 2004-23 2/	24	22	22	22	23	24	48	146
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2004–05	24	22	20	20	21	21	34	87
B2. Export value growth at historical average minus one standard deviation in 2004–05 3/	24	28	36	35	35	35	51	105
B3. US dollar GDP deflator at historical average minus one standard deviation in 2004–05	24	27	31	31	31	31	52	130
B4. Net non-debt creating flows at historical average minus one standard deviation in 2004–05 4/	24	23	20	20	20	20	34	87
B5. Combination of B1-B4 using one-half standard deviation shocks	24	25	28	28	29	28	46	110
B6. One-time 30 percent nominal depreciation relative to the baseline in 2004 5/	24	30	28	28	29	29	48	120
NPV of debt-to-exports ratio								
Baseline	65	59	58	70	83	95	199	336
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2004-23 1/	65	59	49	40	22	0	0	0
A2. New public sector loans on less favorable terms in 2004-23 2/	65	60	60	75	92	109	274	558
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2004–05	65	59	58	70	83	95	199	336
B2. Export value growth at historical average minus one standard deviation in 2004–05 3/	65	106	188	224	265	300	549	758
B3. US dollar GDP deflator at historical average minus one standard deviation in 2004–05	65	59	58	70	83	95	199	336
B4. Net non-debt creating flows at historical average minus one standard deviation in 2004–05 4/	65	61	55	66	79	91	192	331
B5. Combination of B1-B4 using one-half standard deviation shocks	65	72	87	104	124	142	286	460
B6. One-time 30 percent nominal depreciation relative to the baseline in 2004 5/	65	59	58	70	83	95	199	336
Debt service ratio								
Baseline	3	3	3	4	5	6	12	20
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2004-23 1/	3	3	3	4	4	4	1	0
A2. New public sector loans on less favorable terms in 2004-23 2/	3	3	3	4	5	7	15	36
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2004–05	3	3	3	4	5	6	12	23
B2. Export value growth at historical average minus one standard deviation in 2004–05 3/	3	5	6	10	12	14	36	54
B3. US dollar GDP deflator at historical average minus one standard deviation in 2004–05	3	3	3	4	5	6	12	23
B4. Net non-debt creating flows at historical average minus one standard deviation in 2004–05 4/	3	3	3	4	5	6	12	22
B5. Combination of B1-B4 using one-half standard deviation shocks	3	4	4	6	7	9	18	31
B6. One-time 30 percent nominal depreciation relative to the baseline in 2004 5/	3	3	3	4	5	6	12	23
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	45	45	45	45	45	45	45	45

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

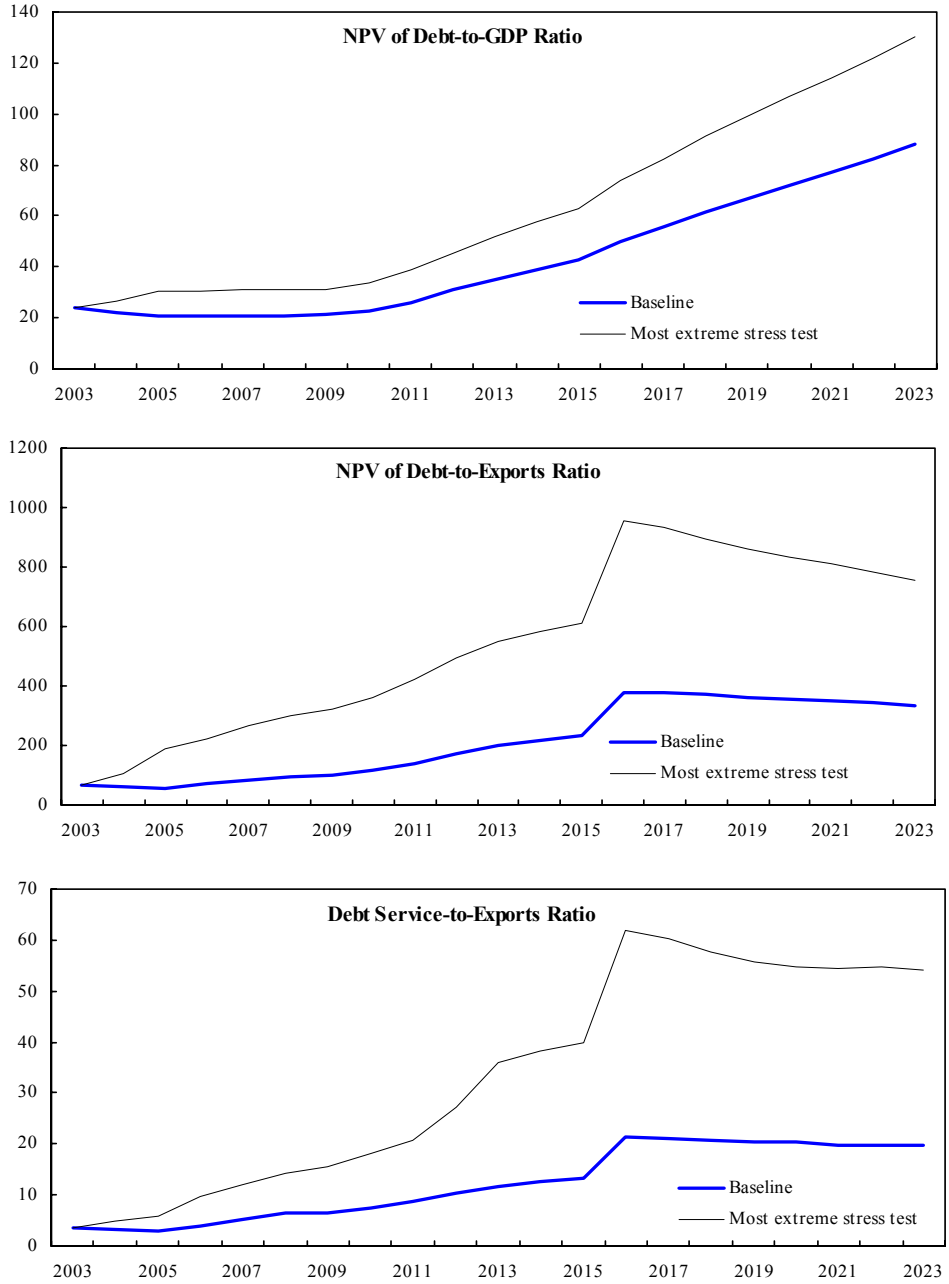
4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 2. Republic of Yemen: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2003–2023

(In percent)



Source: Staff projections and simulations.